$$3x^3 - 4x^2 - 5 = \frac{(3x+1)}{3} \times 3\left(x^2 - \frac{5}{3}x + \frac{5}{9}\right) - \frac{50}{9}$$

$$= (3x+1)\left(x^2 - \frac{5}{3}x + \frac{5}{9}\right) - \left(\frac{50}{9}\right) \qquad \text{(since, } p(x) = d(x)q(x) + r\text{)}$$

Hence the quotient is $\left(x^2 - \frac{5}{3}x + \frac{5}{9}\right)$ and remainder is $\frac{-50}{9}$

Example 3.37 If the quotient on dividing $x^4 + 10x^3 + 35x^2 + 50x + 29$ by $(x+4)$ is $x^3 - ax^2 + bx + 6$, then find the value of a, b and also remainder.

To find the zero of $x+4$;
put $x + 4 = 0$
we get $x = -4$

Solution

Let $p(x) = x^4 + 10x^3 + 35x^2 + 50x + 29$

Standard form $= x^4 + 10x^3 + 35x^2 + 50x + 29$

Coefficient are 1 10 35 50 29

–4	1	10	35	50	29
	0	–4	–24	–44	–24
	1	6	11	6	5 (remainder)

quotient $x^3 + 6x^2 + 11x + 6$ is compared with given quotient $x^3 - ax^2 + bx + 6$

coefficient of x^2 is $6 = -a$ and coefficient of x is $11 = b$

Therefore, $a = -6$, $b = 11$ and remainder $= 5$.

Exercise 3.7

1. Find the quotient and remainder of the following.
   (i) $(4x^3 + 6x^2 - 23x + 18) \div (x+3)$ (ii) $(8y^3 - 16y^2 + 16y - 15) \div (2y-1)$
   (iii) $(8x^3 - 1) \div (2x-1)$ (iv) $(-18z + 14z^2 + 24z^3 + 18) \div (3z+4)$
2. The area of a rectangle is $x^2 + 7x + 12$. If its breadth is $(x+3)$, then find its length.
3. The base of a parallelogram is $(5x+4)$. Find its height, if the area is $25x^2 - 16$.
4. The sum of $(x+5)$ observations is (x^3+125). Find the mean of the observations.
5. Find the quotient and remainder for the following using synthetic division:
   (i) $(x^3 + x^2 - 7x - 3) \div (x - 3)$ (ii) $(x^3 + 2x^2 - x - 4) \div (x + 2)$
   (iii) $(8x^4 - 2x^2 + 6x + 5) \div (4x + 1)$
6. If the quotient obtained on dividing $(8x^4 - 2x^2 + 6x - 7)$ by $(2x + 1)$ is $(4x^3 + px^2 - qx + 3)$, then find p, q and also the remainder.

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